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Migrate roles, users, and

# PolarDB structure and privilege verification tool

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Product

After you migrate Oracle workloads to PolarDB for PostgreSQL (Compatible with Oracle), you must confirm that the destination database matches the Oracle source database in both object structure and user privileges. The PolarDB structure and privilege verification tool compares the object structures and the users, roles, and privileges on both sides, and generates HTML verification reports to help you quickly identify migration differences.

## Scenarios

This tool applies to the following two scenarios:

- After you complete structure migration from Oracle to PolarDB by using Data Transmission Service (DTS), use this tool to compare the object structures of the source and destination databases.
- After you use the migration tool described in [Migrate roles, users and permissions from Oracle to PolarDB for PostgreSQL \(Compatible with Oracle\)](#) to migrate Oracle users, roles, and privileges to PolarDB, use this tool to compare the users and privileges on both sides.

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permissions from Oracle to  
PolarDB for PostgreSQL  
(Compatible with Oracle)

[PolarDB structure and privilege  
verification tool](#)

Migrate self-managed Oracle to

The tool only reads metadata and  
generates reports. It does not  
modify business objects in the  
Oracle source database or the  
PolarDB destination database.

## Scope of application

### Oracle source database requirements

The verification account must have  
at least the `CONNECT` , `CREATE  
SESSION` , and  
`SELECT_CATALOG_ROLE`  
privileges. Example:



```
GRANT CONNECT, CREATE SI
```

### PolarDB destination database requirements

- The destination database is  
PolarDB for PostgreSQL  
(Compatible with Oracle).
- The account that connects to  
PolarDB must have permissions  
to query compatibility views  
such as `DBA_USERS` ,  
`DBA_ROLES` ,  
`DBA_ROLE_PRIVS` ,  
`DBA_SYS_PRIVS` ,  
`DBA_TAB_PRIVS` , and  
`DBA_OBJECTS` .

#### Note

- Before you run  
structure  
verification, use  
Data  
Transmission  
Service (DTS) to  
complete

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structure  
migration from  
Oracle to  
PolarDB.

- Before you run privilege verification, use the migration tool described in [Migrate roles, users, and permissions from Oracle to PolarDB for PostgreSQL \(Compatible with Oracle\)](#) to complete user, role, and privilege migration.

## Runtime environment requirements

Deploy the tool on a machine that can connect to both Oracle and PolarDB at the same time, such as an Elastic Compute Service (ECS) instance. The runtime environment requires Java 1.8 or later.

## Deploy the verification tool

1. Download the verification tool JAR file [polardb-schema-verification-20260827.jar](#), and upload it to the machine that meets the runtime environment requirements described above.
2. Run the following command to check the tool version:



```
java -jar polardb-scl
```

3. Create a `config.json` file with the following content.

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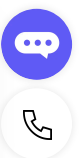


Adjust the database connection strings and the log directory based on your environment. The verification process requires this configuration file.

```
{
  "source": {
    "url": "jdbc:ora:
    "username": "<Or:
    "password": "<Or:
  },
  "destination": {
    "url": "jdbc:poli:
    "username": "<Po:
    "password": "<Po:
  },
  "log": {
    "dir": "/path/to.
  },
  "report": {
    "dir": "/path/to.
  },
  "check": {
    "schemas": ["APP'
  }
}
```

- `source` : the connection to the Oracle source database.
- `destination` : the connection to the PolarDB destination database. Specify the endpoint of the PolarDB cluster.
- `log.dir` : the directory for verification logs.
- `report.dir` : the directory for the generated verification reports.
- `check.schemas` : the list of schemas to verify. If you omit this field, set it to empty, or set it to `"schemas": [ "*" ]` , all schemas are verified.

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## Structure verification

Structure verification assumes that you have used DTS to complete structure migration from Oracle to PolarDB. Before you run the verification, confirm that the `config.json` field in `check.schemas` specifies the intended verification scope.

## Run the verification

Run the following command to compare objects such as tables, views, indexes, sequences, functions, stored procedures, and packages between Oracle and PolarDB:

```
java -jar polardb-schem
```

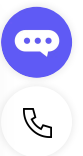
After the command completes, the tool generates the `report.dir` verification report in the `structure_comparison_<database>_<timestamp>.html` directory.

## Interpret the report

The structure verification report includes the following content:

- **Schemas:** compares the schemas of Oracle and PolarDB and shows the **In Both**, **Only in Oracle**, and **Only in PolarDB** states. The built-in schemas of Oracle and PolarDB remain on this tab, so that you can check the system schema differences between the two sides.
- **Object Distribution Summary:** summarizes, by schema and object type, the number of objects that exist on both sides, exist only in Oracle, exist only in PolarDB, and are different.
- **Objects:** shows object details, including the schema, object name, sub-object, object type, comparison result, and reason.

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- **Filter:** filters by schema, object type, and comparison result. You can also hide built-in objects, partitioned objects, materialized view tables, and invalid objects so that objects generated by partitioning or materialization do not obscure real differences.

The comparison states have the following meanings:

- **In Both:** the object or schema exists on both sides.
- **Only in Oracle, Only in PolarDB:** the object or schema exists only on the corresponding side.
- **Different:** the object properties are different.

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## Privilege verification

Privilege verification assumes that you have used the migration tool described in [Migrate roles, users, and permissions from Oracle to PolarDB for PostgreSQL \(Compatible with Oracle\)](#) to complete user, role, and privilege migration. Before you run the verification, confirm that the `config.json` field in `check.schemas` specifies the intended verification scope.

### Run the verification

Run the following command to compare the users, roles, and privileges between Oracle and PolarDB:

```
java -jar polardb-schem
```



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After the command completes, the tool generates the `report.dir` verification report in the `user_privilege_<database>_<timestamp>.html` directory.

### Interpret the report

The privilege verification report includes the following tabs:

- **Users:** compares users and their associated profiles.
- **Roles:** compares roles.
- **Role Privileges:** compares the roles granted to users or roles.
- **System Privileges:** compares system privileges.
- **Table Privileges:** compares object privileges on tables, views, functions, and other objects. The verification scope is controlled by `check.schemas`.

Each tab supports filtering by comparison result and keyword.

Select **Hide Built-in Objects** to hide built-in users, built-in roles, and their associated privilege records.

Previous: Migrate roles, users, and permissions from Oracle to PostgreSQL (Compatible with Oracle) | Next: Migrate self-managed Oracle to PolarDB for PostgreSQL (Compatible with Oracle)

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